

Redistricting Analysis: Rhode Island's 2018 and 2020 Elections

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What is Redistricting?

The redrawing of electoral district boundaries every 10 years based on U.S. Census data.

What is Gerrymandering?

Manipulation of district boundaries to favor one political party or group.

Two common types:

- **Partisan gerrymandering**
- **Racial gerrymandering**

Outlier Analysis in Redistricting

- Purpose: Identify whether a proposed redistricting plan is an outlier compared to a neutral ensemble.
- Methodology:
 - Generate a large ensemble of valid plans using tools like GerryChain.
 - Compare metrics (e.g., partisan bias, compactness) of the proposed plan against the ensemble.
- Significance:
 - Provides empirical evidence of potential gerrymandering.
 - Supports legal and policy decisions.

What is ReCom?

A family of Markov chains for redistricting

- Overview: ReCom is a Markov chain algorithm designed to sample redistricting plans by recombining adjacent districts.
- Process:
 1. Select two adjacent districts.
 2. Merge them into a single unit.
 3. Generate a random spanning tree over the merged unit.
 4. Split the tree into two new districts with balanced populations.
- Advantages:
 - Maintains contiguity and population balance.
 - Efficiently explores the space of valid plans.
 - Produces more substantial changes between steps compared to single-node flip methods.

Analysis and Applications of ReCom

- Efficiency: Demonstrated faster mixing times, leading to more diverse plan samples.
- Tends to produce plans with better compactness and adherence to traditional redistricting principles.
- Applied to assess the fairness of proposed maps.
- Used in legal contexts to provide evidence of gerrymandering.
- Extensions:
 - RevReCom: A reversible variant ensuring convergence to a known stationary distribution.
 - Metric-Biased Sampling: Incorporates specific criteria (e.g., community preservation) into the sampling process.

Background of RI

Politics and Demographics

Racial Demographics:

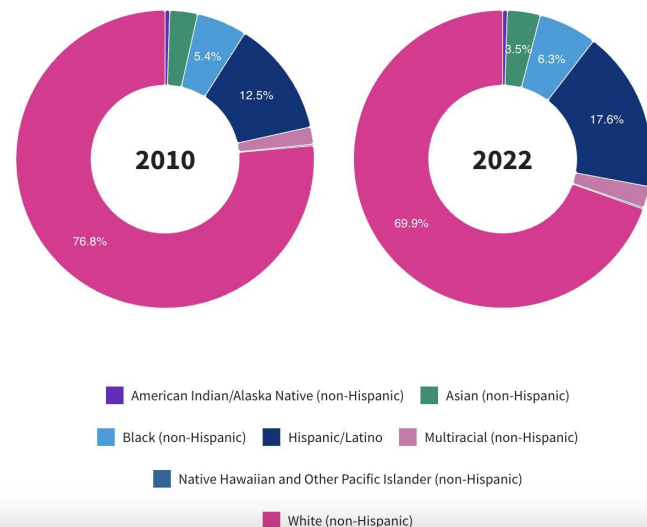
- White (Non-Hispanic): 69–72%
- Hispanic/Latinx: 16–18%
- Black: 5–9%; Asian: 3–4%
- Communities of color growing

Political Profile:

- Strong Democratic state since 1928.
- 2024 House makeup: 64 Democrats, 10 Republicans, 1 Independent.

Population Overview:

- ~1.1 million (2023–2024), second-highest population density in the U.S.



Sample of Racial Makeup of RI (USAFacts, 2025)

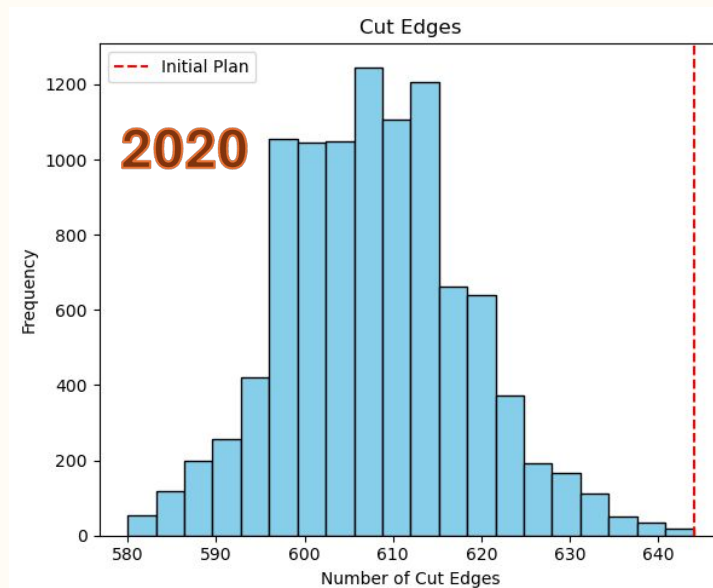
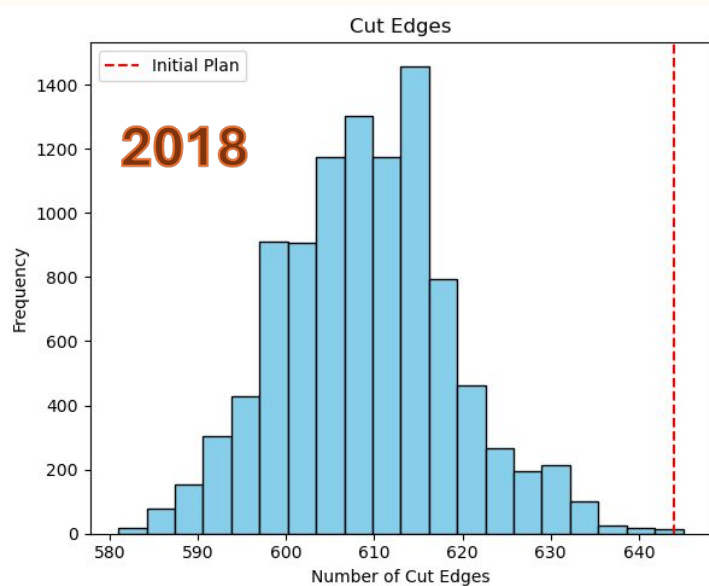
Gerry Chain Results

Cut Edges

Cut Edges is usually used to measure the compactness of constituencies. Fewer cut edges means smoother and more compact constituency boundaries.

The cut edges of the initial plan are significantly higher than most simulation plans, which means that its compactness may be poor.

Similar to 2018.



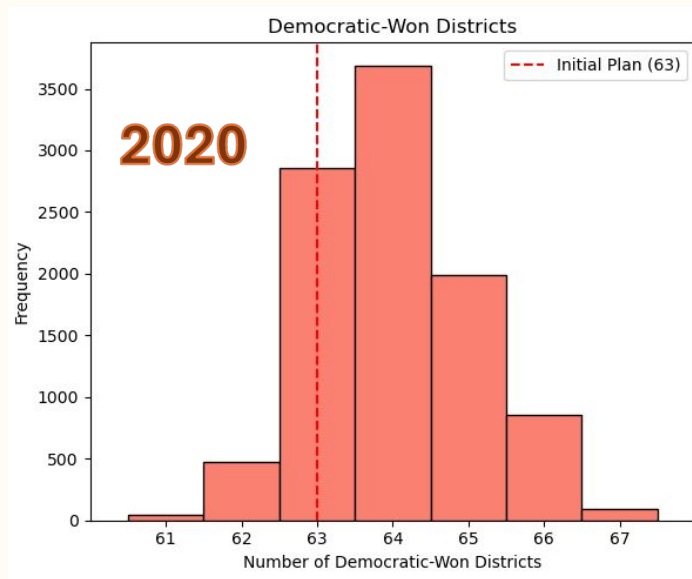
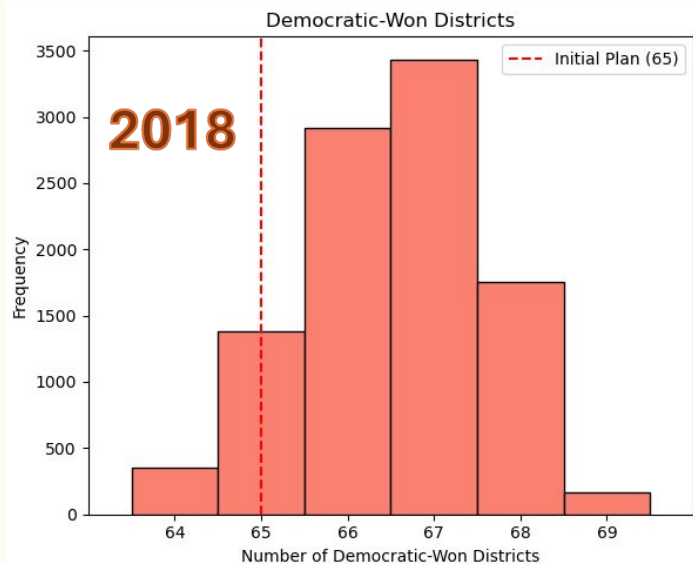
Gerry Chain Results

Democratic-won

Based on historical election data, the number of constituencies won by the Democratic Party is calculated in each plan.

The number of Democratic wins in the initial plan is slightly lower than that in most simulation plans, and there may be a slight Republican tilt (less favorable to Democrats).

The initial plan is on the left side of the simulation distribution, showing a slight deviation; but it is not seriously skewed.



Gerry Chain Results

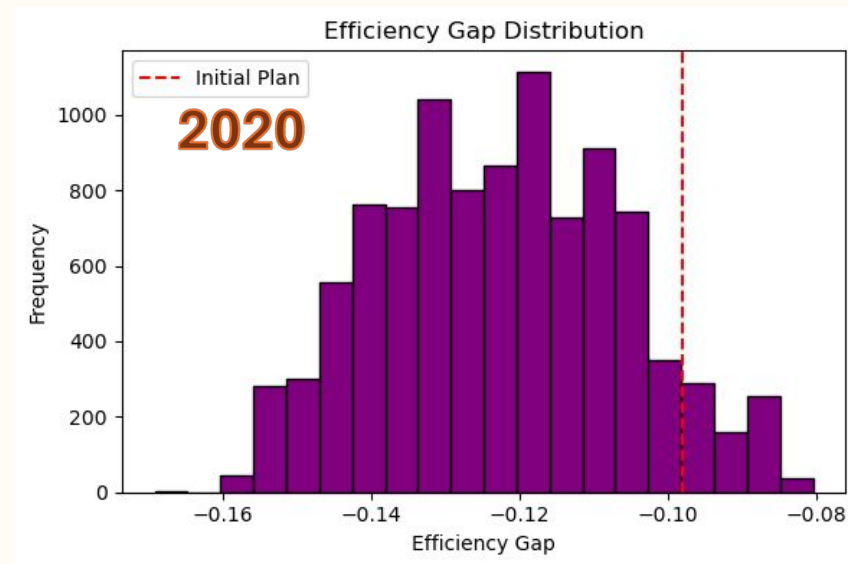
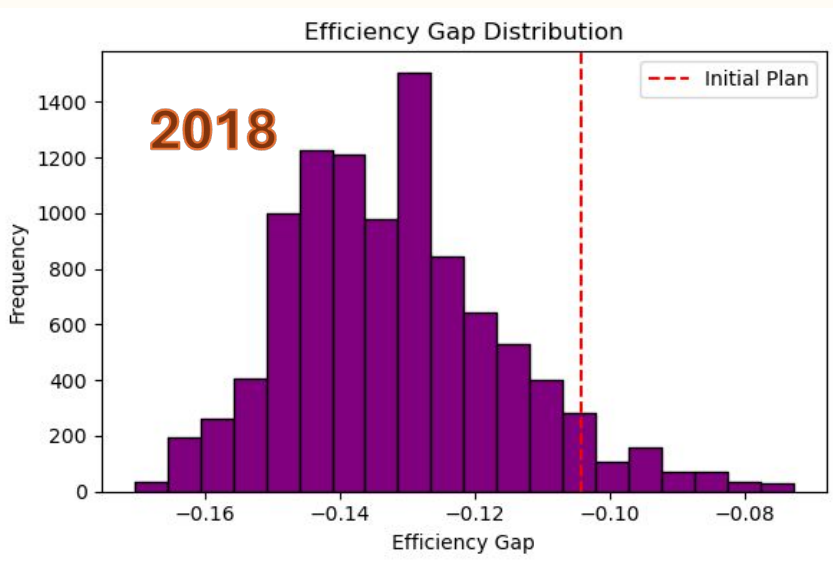
Efficiency Gap

Efficiency Gap measures the difference in "wasted votes" between the two parties and is used to identify whether there is a systematic partisan advantage.

Positive values are biased towards the Democratic Party, and negative values are biased towards the Republican Party.

Suggests that the initial proposal is slightly less favorable to Republicans than the simulations.

Similar to 2018.



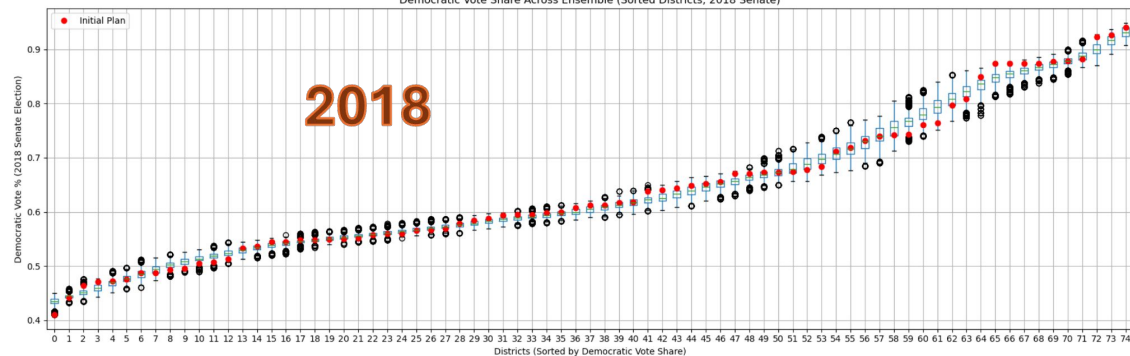
Gerry Chain Results

Marginal Box Plot

To show the distribution of the Democratic vote ratio in each sorted constituency under different scenarios (median, upper and lower quartiles, extreme values), as well as the position of the initial scenario (red dots).

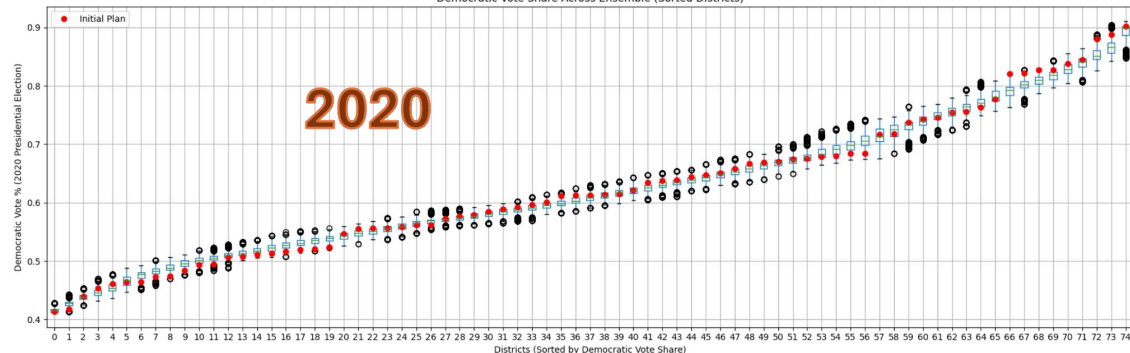
2018 Senate Election - Marginal Box Plot

Democratic Vote Share Across Ensemble (Sorted Districts, 2018 Senate)



2020 Presidential Election - Marginal Box Plot

Democratic Vote Share Across Ensemble (Sorted Districts)



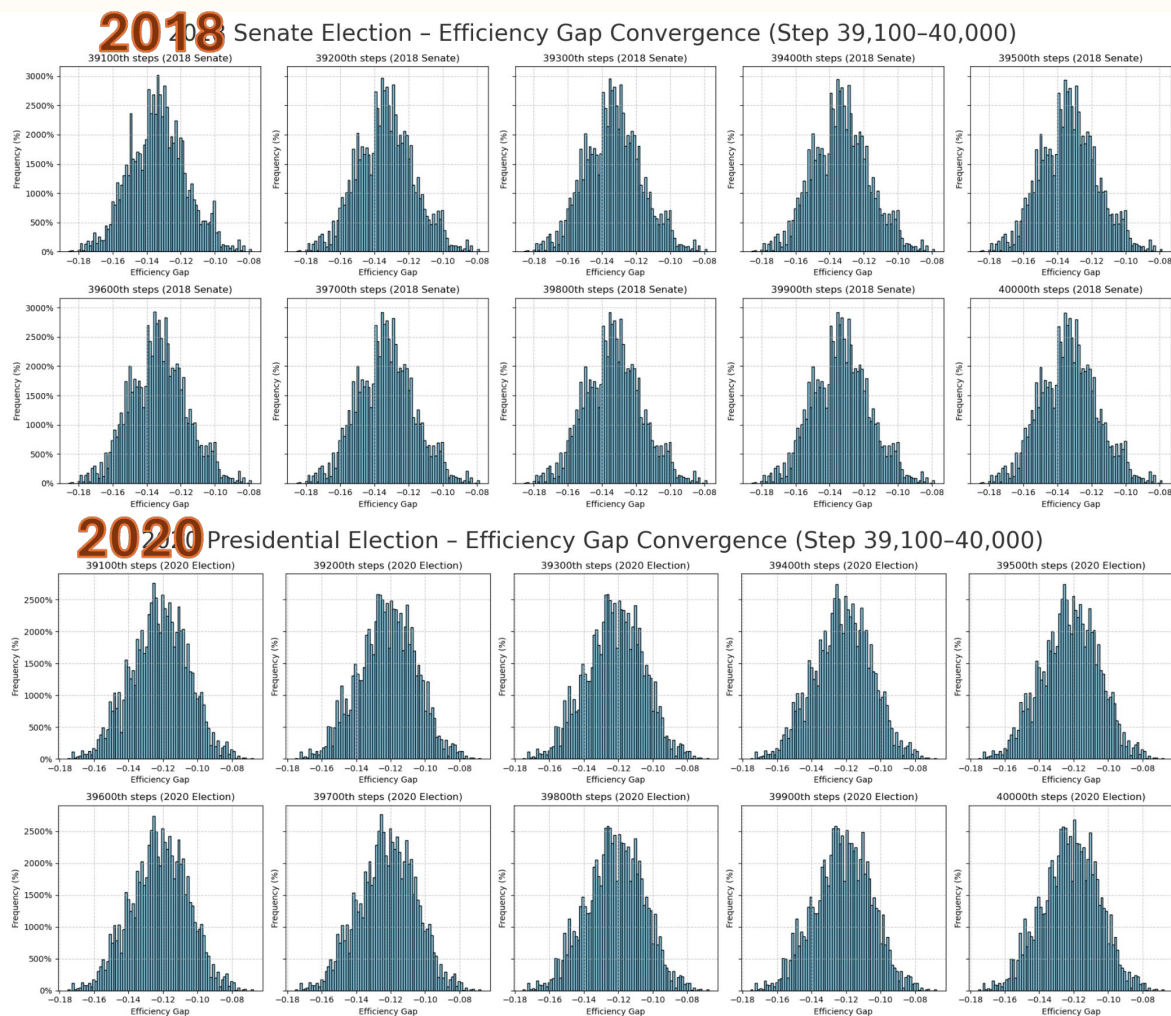
Box plots indicate that the initial plans for both elections generally fall within the ensemble distribution, but the 2018 plan shows slightly more variation away from neutral expectations compared to the 2020 plan.

Gerry Chain Results

Convergence

Markov Chain Monte Carlo simulations explore the space of possible redistricting plans. Convergence means the sampled plans have stabilized, and the metrics (e.g., efficiency gap) stop shifting significantly over time.

The distributions of efficiency gap remain consistent in the final 1,000 steps for both elections, confirming that the Markov Chain has converged. This validates the reliability of the ensemble-based analysis.



Gerry Chain Results

Conclusion

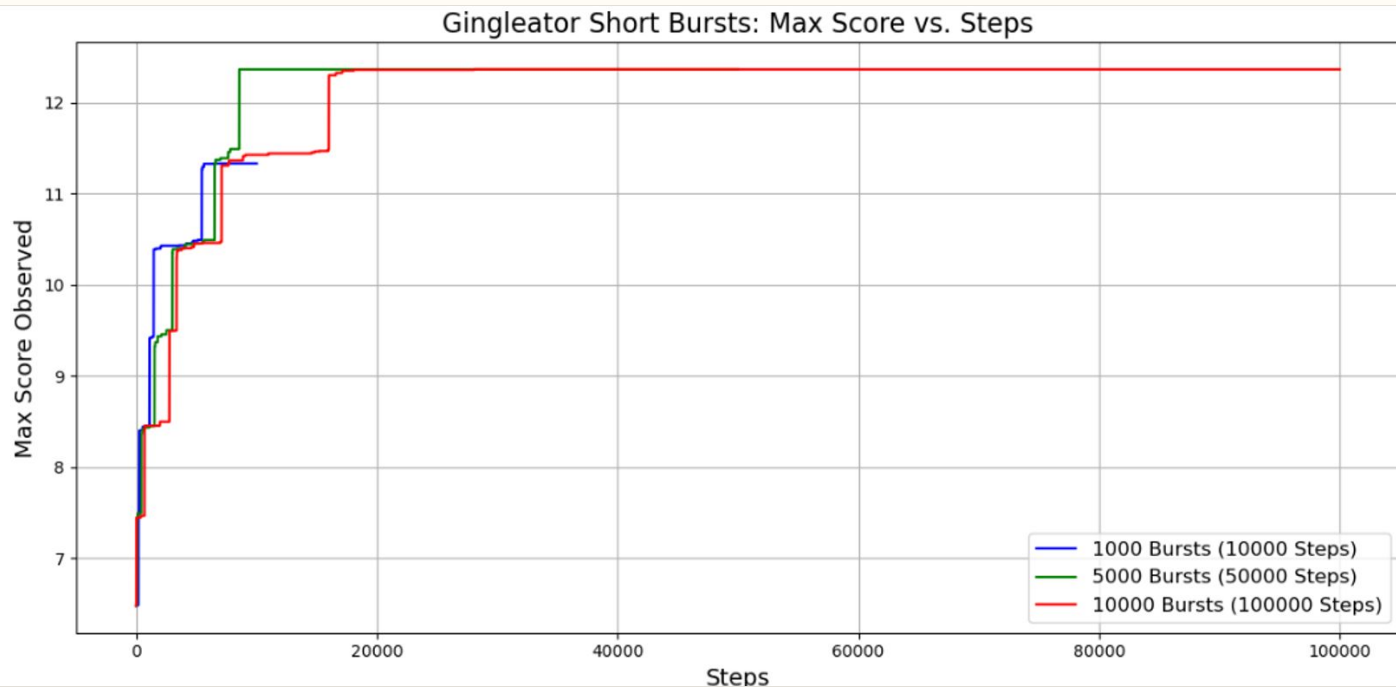
- **Had more cut edges** than most ensemble plans → less compact.
- **Won fewer Democratic districts** than neutral alternatives → slight partisan skew.
- **Showed smaller efficiency gaps** than average → no aggressive gerrymandering.
- **Converged well** over 40,000 steps → results are statistically reliable.

Short Burst

- We used the Gingleator class from gerrychain to do the shortburst.
- We used the function `reward_partial_dist` with default value.
- classmethod : `reward_partial_dist` (part: Partition, minority_perc_col: str, threshold: float) → float, which sets the threshold to 50%.
- `reward_partial_dist`: Given a Partition, this function will return $n + \max(\{p_i : p_i < t\})$
- Here n is the number of districts that meet the requirements, and $\max p_i$ is the percentage of the district that does not meet requirements but has the largest percentage of Latino voting age population.

Short Burst

Here we run three 10 steps bursts for 10,000, 20,000, 100,000 steps. We find that the shortburst algorithm converges to a local maximum of Latino-majority districts within 5,000 bursts. But we do not know how to prove this is also the global maximum since we do not have a good algorithm or a quantum computer.





Questions?



Thank You